Solving Quadratic Equations: Choose a Method, Read the Roots

Algebra 1

Focus: Choosing among square roots, factoring, and the quadratic formula, and interpreting roots in context.

Before you solve, choose. These problems are mixed on purpose.

- **Square roots** — when the variable appears only inside one squared quantity and there is no linear term: $ax^2 = k$ or $a(x-h)^2 = k$.
- **Factoring** — when the quadratic factors over the integers (always check for a common factor first).
- **Quadratic formula** — when it does not factor, or when you cannot tell quickly. It always works.

Write the method you picked on the **Method** line, then solve. In the word problems, finish by saying what the roots mean for the situation.

1. Solve: $x^2 = 49$

Method: ________________________

Answer: ________________

2. Solve: $x^2 - 5x + 6 = 0$

Method: ________________________

Answer: ________________

3. Solve, leaving your answers in exact radical form: $x^2 + 4x + 1 = 0$
Method: ____________________

Answer: ______________

4. Solve: $2(x-3)^2 = 32$
Method: ____________________

Answer: ______________

5. A ball is thrown straight up from ground level. Its height in feet after t seconds is $h = -16t^2 + 48t$. Solve $h = 0$ to find every time the ball is at ground level, then say how long the ball is in the air.
Method: ____________________

Answer: ______________ seconds

6. Solve: $3x^2 = 12x$
Method: ______________________

Answer: ______________

7. Solve: $2x^2 - 3x - 5 = 0$
Method: ______________________

Answer: ______________

8. A rectangular garden is 5 metres longer than it is wide. Let w be the width in metres, so the area is $w(w+5) = 84$ square metres. Find both roots of $w^2 + 5w - 84 = 0$, then state the width of the garden.
Method: ______________________

Answer: ______________ m

9. Solve: $3(x+2)^2 = 27$
Method: ________________________

Answer: __________________

10. Solve: $6x^2 + 7x - 3 = 0$
Method: ________________________

Answer: __________________

11. A ball leaves a player's hand 5 feet above the ground moving upward at 40 feet per second. Its height in feet after t seconds is $h = -16t^2 + 40t + 5$. Find the time when the ball lands ($h = 0$), rounded to the nearest hundredth of a second, and explain why the other root is thrown out.
Method: ________________________

Answer: __________________ seconds

12. Synthesis. A club's profit in dollars from selling n sweatshirts is $P = -n^2 + 30n - 189$.
Method: ______________________

(a) Break-even means $P = 0$. Solve $-n^2 + 30n - 189 = 0$ for n.

Answer: ________________

(b) The manager says selling 20 sweatshirts breaks even. Use your roots to say whether she is right, and describe what happens to the profit for numbers of sweatshirts between the two break-even values.

Answer: ________________